Digital Document Archive

Requirements Specification — v0.5

System	Digital archive for a single-owner business — GoBD system of record for born-digital documents
Scope	**Store & retrieve only.** The archive does not create invoices; outgoing invoices are created in the accounting program and fed in as copies.
Primary user	Owner (sole user)
Jurisdiction	Germany — GoBD applies; 10-year retention
Storage decision	Cloud, GoBD-compliant DMS TEST
Status	Provisionally Ready — every TEST value needs owner / tax-advisor confirmation

> **Provenance note.** This specification was produced through an AI-assisted requirements workflow (elicitation → drafting → review → test cases). Items marked TEST are placeholder answers invented to unblock the workflow — they are **not** real stakeholder decisions and must be confirmed by the owner, and by the tax advisor for GoBD and privacy items. A human owns every final call.

1. Business Goals

ID	Goal
BG-1	**Findability** — one location to find every document.
BG-2	**Audit-readiness** — pull bank documents, bills, and invoices as PDF on demand; keep for 10 years.
BG-3	**Repeatable filing** — the owner knows exactly what to save where.

2. Assumptions

- **A-1** Windows environment (desktop + laptop). **A-2** Documents are PDF or scans (scanner outputs PDF); no photos, no Word files.
- **A-3** The accounting / "booking" program is out of scope to change. **A-4** Documents and party names are in German.
- **A-5** Migration covers the last 3 years only — the full life of the 3-year-old business, so migration is complete.
- **A-6** Volume is low: peak ~50 documents/month (Jan–Mar), a few hundred/year, well under 5 GB over 3 years.

3. Decisions Recorded

#	Decision
D-1	GoBD applies — audit-proof/unalterable storage + Verfahrensdokumentation required.

D-2	Storage cost ceiling €10/month; met by a low-cost GoBD-compliant cloud DMS (~€9/mo). TEST
D-3	Accounting program offers export/API — weekly scheduled PDF batch.
D-4	Retention clock = end of calendar year of the last entry (Fiscal Year).
D-5	Backup: weekly, to one portable hard disk.
D-6	Volume: outgoing ≤20/mo (usu.10); incoming usu.5, peak 30/mo Jan–Mar.
D-7	Acceptable search response ≤ 60 seconds.
D-8	Private-document exclusion is manual, by the owner.
D-9	Email attachments: one-off triage list only.
D-10	Category set extendable anytime; required fields Date, Category, Counterparty, Fiscal Year; optional Description, Amount, Physical Location.
D-11	Laptop out of scope — owner consolidates onto the desktop.
D-12	Metadata entered manually by the owner (no OCR).
D-13	Keep paper originals — scans are convenience copies. TEST advisor accepts.

4. Functional Requirements

A. Capture & Ingestion

ID	The system shall…
FR-1	allow the owner to add a document by uploading a PDF or scanned file (max **50 MB** TEST; malware-scanned before it becomes available).
FR-2	support a one-time bulk import of the last 3 years of documents from the **desktop PC**.
FR-3	receive a **weekly PDF batch** of newly issued outgoing invoices from the accounting program.
FR-4	allow the owner to add a physical document by scanning it to PDF and importing it.
FR-5	produce a one-off list of email attachments (sender, date received) to support triage during migration.
FR-25	perform a one-time bulk export of **all outgoing invoices for the last 3 years** from the accounting program, with a **reconciliation count** against the accounting tool.
FR-29	archive an emailed incoming bill when the owner saves the attachment and captures it via FR-1 (ongoing routine).

B. Classification & Metadata

ID	The system shall…
FR-6	store per document — required: **Date, Category, Counterparty, Fiscal Year**; optional: Description, Amount, Physical Location. Entered manually. "Counterparty" is defined per category in the Verfahrensdokumentation.
FR-7	organise documents under a category set (Outgoing Invoice, Incoming Bill, Contract, Bank Document, Insurance) and allow the owner to **add categories at any time**.
FR-8	hold business documents only; the owner excludes private documents before import (process step).

C. Search & Retrieval

ID	The system shall…
FR-9	let the owner search by counterparty name.
FR-10	let the owner filter by category.
FR-11	let the owner filter by document-date range.
FR-12	let the owner filter by amount range.
FR-13	let the owner open/download the original document as PDF.
FR-14	let the owner select and export a set of documents as PDF for a tax check.

D. Retention & Records

ID	The system shall…
FR-15	retain every document for **10 years from the end of the calendar year of its Fiscal Year** and keep it openable as PDF for that period.
~~**FR-16**~~	~~(retired — subsumed by FR-20)~~
FR-23	index every document so it is individually retrievable, recording its source and capture date.
FR-24	store each document in its original received format without altering content (byte-identical).

E. GoBD Compliance

ID	The system shall…
FR-20	store documents in an **audit-proof (WORM) GoBD-compliant cloud DMS** so a stored document cannot be altered or deleted before retention expiry. Applies as system-of-record to **born-digital** documents; scans of retained paper are convenience copies at a documented lower bar.
FR-21	maintain an **append-only audit log** (capture, export, correction), retained 10 years and included in the backup.
FR-22	be accompanied by a **Verfahrensdokumentation** with all required sections and a tax-advisor sign-off (AI drafts → owner amends → advisor reviews).
FR-27	provide a GoBD-compliant **void/annotate** path (reason + audit entry, original retained) and a distinct **GDPR erasure** path for private documents captured in error.

F. Storage, Backup & Security

ID	The system shall…
FR-17	write a **weekly backup** to one portable hard disk.
FR-18	use a configurable storage location — **decided: cloud** TEST.
FR-26	perform a documented **restore test annually**, result recorded in the Verfahrensdokumentation.
FR-28	encrypt documents **at rest** — primary cloud store and backup disk (AES-256 TEST).

G. Access

ID	The system shall…

FR-19	restrict access to the owner via authentication (single user).

5. Non-Functional Requirements

ID	Requirement (measurable)
NFR-1	Durability — no loss of documents older than the last successful weekly backup.
NFR-2	Backup — weekly to one portable disk; **RPO ≤ 7 days**; owner accepts the exposure TEST.
NFR-3	Search/filter returns results in **≤ 60 s** at expected volume.
NFR-4	Capacity — a few hundred documents/year, under 5 GB over 3 years, without redesign.
NFR-5	Usability — the owner can file and retrieve a document **unaided using the written instructions**.
NFR-6	Compliance — meets GoBD: unalterable storage, complete/traceable capture, retrievability throughout retention, and Verfahrensdokumentation.

6. Privacy (GDPR)

- **Personal data:** outgoing invoices and contracts carry customer / counterparty personal data.
- **Lawful basis:** statutory tax-retention obligation (§147 AO / GoBD) TEST; recorded in the processing register (Verzeichnis von Verarbeitungstätigkeiten).
- **Erasure:** refused for tax-retained documents; private documents captured in error are erasable via FR-27.
- **At rest:** encrypted per FR-28; backup media stored safely per the Verfahrensdokumentation.

7. User Stories & Acceptance Criteria

US-1 — File a document with the right metadata *(BG-1, BG-3 · FR-1, FR-6, FR-29)*

- **AC-1.1** Given a scanned PDF of an incoming bill, when I import it with category, counterparty, date, fiscal year, then it is stored and found by that counterparty.
- **AC-1.2** Given a required field is missing, when I save, then the system blocks it and names the missing field.
- **AC-1.3** Given a non-PDF file, when I import, then it is rejected with a clear message.
- **AC-1.4 / 1.5** Given a PDF at/over the 50 MB limit, when I save, then it is accepted / rejected with the limit stated.
- **AC-1.6** Given a file fails the malware scan, when captured, then it is not archived, access is blocked, and the failed scan is recorded.
- **AC-29.1** Given an emailed bill saved from the inbox, when captured via FR-1, then it is archived like any incoming bill.

US-2 — Bulk-migrate the last 3 years *(BG-1 · FR-2, FR-25)*

- **AC-2.1** Given the consolidated desktop folders, when I import, then documents are stored with source and category recorded.
- **AC-2.2** Given documents older than 3 years, when I import, then they are excluded.
- **AC-25.1** Given 3 years of outgoing invoices in the accounting program, when migration runs, then every one is archived with customer, date, amount.
- **AC-25.2 / 25.3** Given migration completes, when the archived count equals / differs from the accounting-tool count, then it is marked Complete / Incomplete with missing invoices listed.

US-3 — Auto-capture new outgoing invoices *(BG-1, BG-2 · FR-3)*

- **AC-3.1** Given an invoice is finalised, when the weekly batch runs, then a PDF copy is archived with customer, date, amount.
- **AC-3.2** Given the feed fails, when the transfer fails, then it is recorded and surfaced — no silent loss.
- **AC-3.3** Given the same invoice is delivered twice, when received, then only one copy is stored.

US-4 — Retrieve for a tax audit *(BG-2 · FR-13, FR-14)*

- **AC-4.1** Given a tax check for a year, when I filter by that range and select Bank Document, Incoming Bill, Outgoing Invoice, then I can export the matching documents as PDF.
- **AC-4.2** Given no documents match, when I export, then a clear empty result is shown and no error.

US-5 — Search the way I think *(BG-1 · FR-9–FR-12)*

- **AC-5.1** Given bills from a supplier, when I search that name, then all their documents are returned.
- **AC-5.2** Given I filter by amount range and year, when applied, then only documents matching both are returned.
- **AC-5.3** Given no documents match, when I search, then a clear empty-state message is shown and no error.

US-6 — Keep documents safe for 10 years *(BG-2 · FR-15, FR-17, FR-26, FR-28)*

- **AC-6.1** Given a stored document within its 10-year window, when I attempt to delete it, then deletion is refused.
- **AC-6.2** Given the primary store is lost, when I restore from backup, then all documents and metadata are recovered.
- **AC-26.1 / 26.2** Given a backup exists, when the annual restore test runs, then all documents and metadata are recovered and recorded / a failure is recorded and the owner alerted.
- **AC-28.1–28.3** Given documents at rest / a backup disk, then the primary store and backup are encrypted, and the disk is unreadable without the key.

US-7 — Nothing can be tampered with *(BG-2 · FR-20, FR-21, FR-22, FR-23, FR-24, FR-27)*

- **AC-7.1** Given an archived document, when I attempt to edit or delete it before retention expiry, then it is prevented.
- **AC-7.2** Given a metadata error, when I correct it, then the change is recorded as a tracked change with a timestamp (original preserved).
- **AC-7.3** Given any capture or export, when it happens, then it is written to the append-only audit log.
- **AC-27.1–27.4** Given a wrongly captured document, when I void it with a reason, then it is marked Void (excluded from search and tax export), the original retained, an audit entry written; a void without a reason is blocked; a private document captured in error is erased via the GDPR path.
- **AC-22.1–22.3** Given the solution is delivered, then the Verfahrensdokumentation contains all required sections and a tax-advisor sign-off; adding a category updates the indexing section and change history.
- **AC-23.1** Given any archived document, when retrieved, then it shows its source and capture date and is individually addressable.
- **AC-24.1** Given a captured PDF, when compared to the archived copy, then the file is byte-identical (hash match).

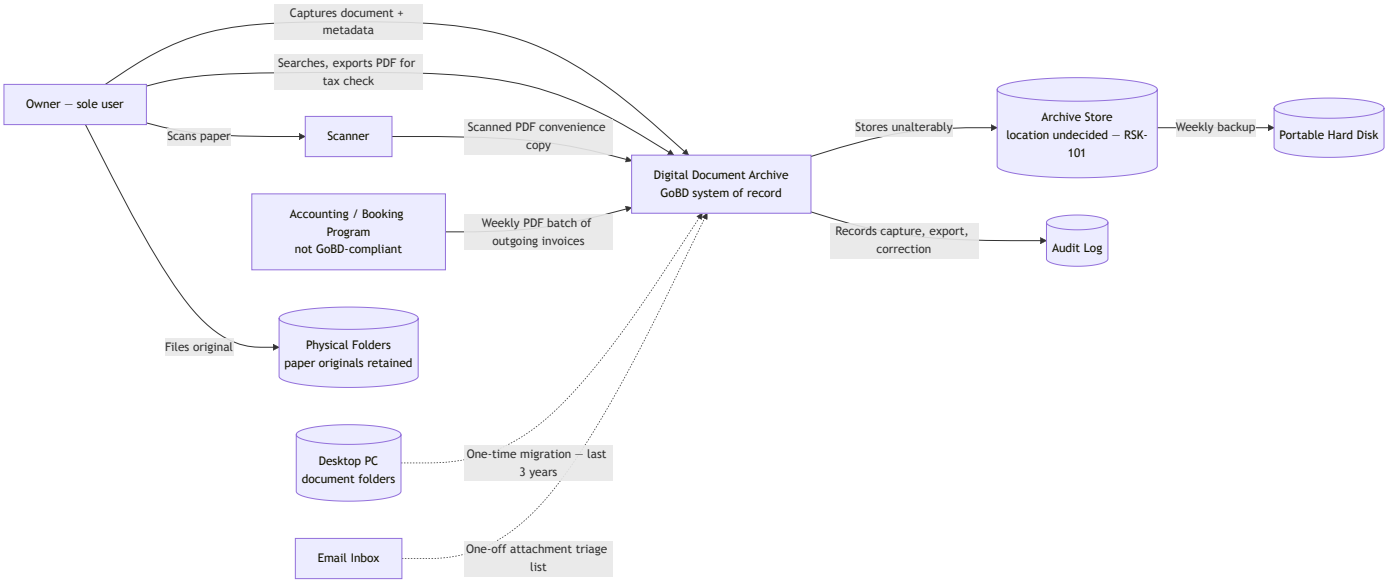
US-8 — Only the owner gets in *(new · FR-19)*

- **AC-19.1 / 19.2** Given valid / invalid credentials, when signing in, then access is granted / denied with a non-specific message.
- **AC-19.3** Given a session idle beyond 15 min TEST, when the owner acts, then re-authentication is required.
- **AC-19.4** Given 5 consecutive failed attempts TEST, when a 6th is made, then sign-in is locked for a cool-off period.

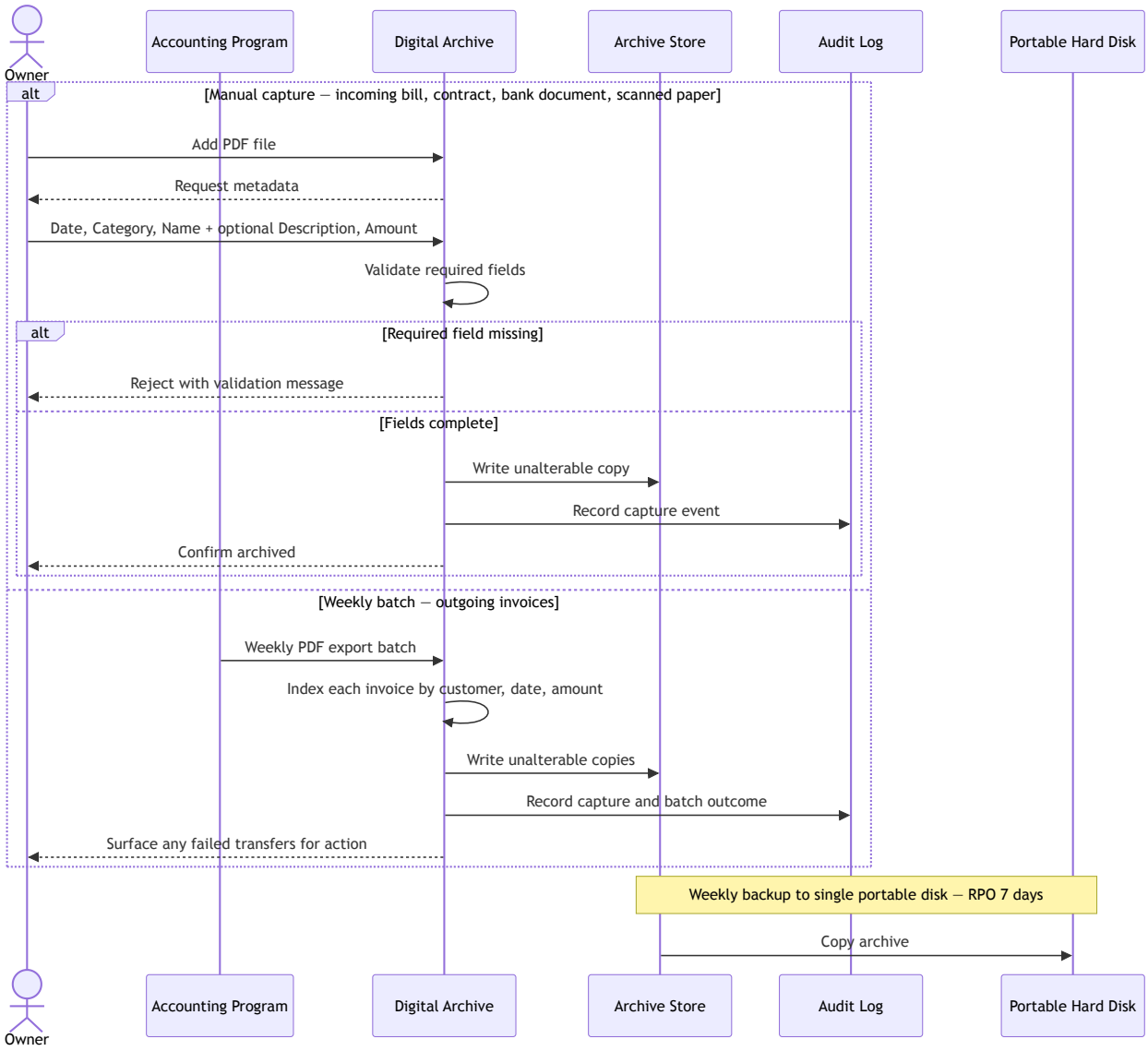
8. Diagrams

All three machine-verified with mermaid-cli. Sources in `digital-archive-diagrams/`.

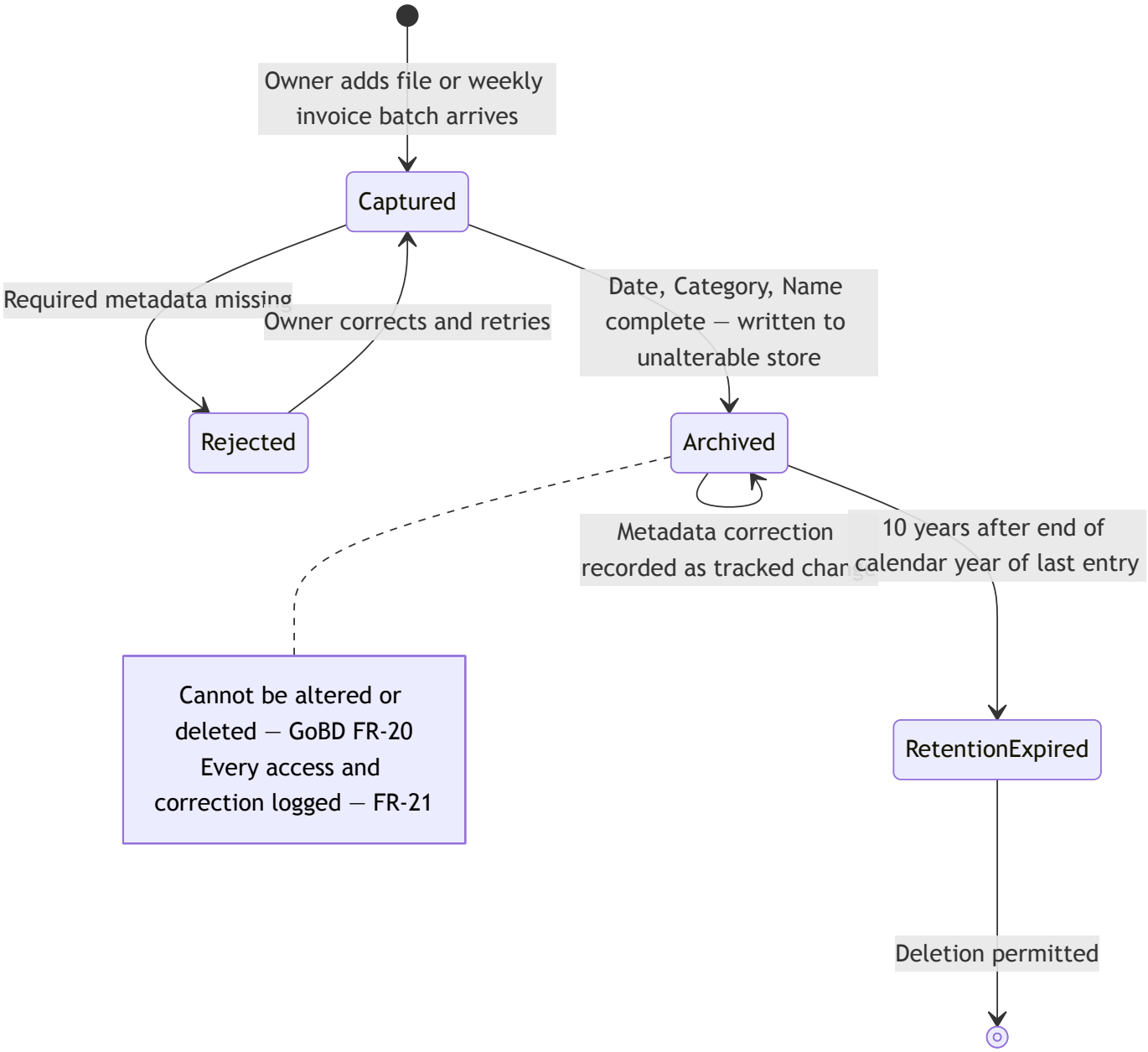
8.1 Context diagram



8.2 Ingestion sequence



8.3 Document lifecycle



9. Test Cases

55 cases; each traces to an acceptance criterion. Test data values are illustrative.

Test ID	AC	Title	Preconditions	Steps	Test data	Expected result	Type
TC-01-01	AC-1.1	Capture incoming bill with metadata	Owner authenticated	Add PDF; set Category, Counterparty, Date, Fiscal Year; save	Incoming Bill; Stadtwerke München; 2026-04-12; FY2026	Stored; found by counterparty	Positive
TC-01-02	AC-1.2	Reject missing Category	Owner authenticated	Add PDF; leave Category empty; save	Category empty	Blocked; names Category	Negative
TC-01-03	AC-1.2	Reject missing Counterparty	Owner authenticated	Add PDF; leave Counterparty empty; save	Counterparty empty	Blocked; names Counterparty	Negative
TC-01-04	AC-1.2	Optional fields may be omitted	Owner authenticated	Fill only 4 required fields; save	Desc/Amount/Location omitted	Saved	Boundary
TC-01-05	AC-1.3	Reject non-PDF	Owner authenticated	Add file; save	bill.docx	Rejected; format message	Negative
TC-01-06	AC-1.1	German umlauts stored & searchable	Owner authenticated	Capture umlaut counterparty; search it	Müller & Söhne GmbH	Stored unchanged; returned	Data edge
TC-01-07	AC-1.1	Duplicate warning on repeat	Identical bill archived	Capture same Counterparty+Date+Amount; save	duplicate	Warning before commit	Negative
TC-01-08	AC-1.4	Accept file at max size	Owner authenticated	Add PDF at limit; save	50.0 MB	Accepted	Boundary
TC-01-09	AC-1.5	Reject oversized file	Owner authenticated	Add PDF above limit; save	50.1 MB	Rejected; limit stated	Boundary
TC-01-10	AC-1.6	Block malware-flagged file	Scanner returns positive	Add file; save	EICAR test file	Not archived; blocked; recorded	Negative
TC-02-01	AC-2.1	Migrate desktop folders	Consolidated folders present	Run migration; inspect	bills/Bank/Contracts/Insurance	Imported; source+category recorded	Positive
TC-02-02	AC-2.2	Exclude >3-year docs	4-year-old doc present	Run migration; search it	dated 2022-01-15	Not imported	Boundary
TC-02-03	AC-2.2	Include doc at 3-year edge	Doc at cutoff present	Run migration; search it	at cutoff date	Imported	Boundary
TC-03-01	AC-3.1	Weekly batch archives invoice	Invoice finalised; batch due	Run batch; open archive	RE-2026-0141	PDF stored; customer/date/amount	Positive
TC-03-02	AC-3.2	Failed transfer surfaced	Export endpoint down	Run batch; check report	API error	Recorded; surfaced; no silent loss	Negative
TC-03-03	AC-3.3	Re-run does not duplicate	Batch already ran	Re-run; count copies	invoice twice	One copy	Negative

TC-03-04	AC-3.1	Empty batch completes	No invoices that week	Run batch	zero invoices	Completes; logged; no error	Data edge
TC-04-01	AC-4.1	Export audit set as PDF	Docs across categories	Filter year+3 categories; export	FY2025	Matching docs exported as PDF	Positive
TC-04-02	AC-4.2	Export with no matches	Empty range	Filter empty range; export	FY2019	Empty-result message; no partial file	Negative
TC-05-01	AC-5.1	Search by counterparty	3 bills one supplier	Search name	Stadtwerke München	All 3 returned	Positive
TC-05-02	AC-5.2	Combined amount+date filter	Varied docs	Filter €500–1,000 + year	€500–1,000	Only both-match returned	Positive
TC-05-03	AC-5.2	Amount boundary inclusivity	Docs at €500 & €1,000	Apply €500–1,000	499.99/500/1000/1000.01	Bounds included; outside excluded	Boundary
TC-05-04	AC-5.3	Search no results	Archive populated	Search non-matching term	zzzz-nonexistent	Empty-state; no error	Negative
TC-05-05	AC-5.1	Special characters safe	Archive populated	Search punctuation/symbols	& % " long string	Handled safely; no crash	Data edge
TC-06-01	AC-6.1	Delete blocked in retention	Doc FY2026	Attempt delete	window open	Refused with reason	Negative
TC-06-02	AC-6.1	Boundary: day before expiry	Doc FY2015; clock 2025-12-31	Attempt delete	EOY+10y	Still refused	Boundary
TC-06-03	AC-6.1	Boundary: after expiry	Doc FY2015; clock 2026-01-01	Attempt delete	elapsed	Deletion permitted	Boundary
TC-06-04	AC-6.1	Fiscal-year edge 31Dec/1Jan	Two docs either side of EOY	Compare retention dates	2026-12-31 / 2027-01-01	Differ by exactly one year	Boundary
TC-06-05	AC-6.2	Restore recovers docs+metadata	Backup taken; primary cleared	Restore from disk; verify	full archive	All docs+metadata recovered	Positive
TC-07-01	AC-7.1	Archived doc cannot be edited	Doc archived	Attempt edit/replace	any PDF	Prevented; logged	Negative
TC-07-02	AC-7.2	Correction as tracked change	Doc with wrong counterparty	Correct; inspect history	Müler→Müller	Recorded w/ timestamp; original kept	Positive
TC-07-03	AC-7.3	Capture+export logged	Archive populated	Capture; export; read log	any	Both events logged w/ timestamp	Positive
TC-07-04	AC-7.3	Audit log append-only	Entries exist	Attempt modify/delete entry	any entry	Refused	Negative
TC-19-01	AC-19.1	Sign in valid	Account exists	Enter valid; submit	valid login	Access granted	Positive
TC-19-	AC-19.2	Reject invalid	Account exists	Wrong password; submit	wrong password	Denied; non-specific message	Negative

02							
TC-19-03	AC-19.3	Session expiry	Signed in	Idle 15 min; act	idle 15 min	Re-auth required	Boundary
TC-19-04	AC-19.4	Lockout after 5 fails	Account exists	Fail 5×; attempt 6th	6 wrong	6th locked for cool-off	Negative
TC-25-01	AC-25.1	Migrate 3y outgoing invoices	Accounting holds 3y	Run migration; inspect	500 invoices	All archived w/ customer/date/amount	Positive
TC-25-02	AC-25.2	Reconciliation match=Complete	Counts equal	Compare counts	500=500	Marked Complete	Positive
TC-25-03	AC-25.3	Mismatch lists gaps	2 dropped	Compare; view report	498 vs 500	Incomplete; 2 listed	Negative
TC-26-01	AC-26.1	Annual restore test passes	Backup exists	Restore to test loc; verify	full archive	All recovered; recorded	Positive
TC-26-02	AC-26.2	Failed restore alerts	Backup corrupted	Run restore test	unreadable backup	Recorded Failed; owner alerted	Negative
TC-27-01	AC-27.1	Void with reason	Wrongly captured doc	Void w/ reason; search+audit	reason=duplicate	Void; excluded; original kept; audited	Positive
TC-27-02	AC-27.2	Void without reason blocked	Doc archived	Void, empty reason	reason empty	Blocked	Negative
TC-27-03	AC-27.3	Private-doc erasure	Private doc in error	Run erasure; check audit	shopping bill	Removed; erasure recorded; ≠ void	Positive
TC-27-04	AC-27.4	Voided excluded from export	Voided doc in period	Run tax export	voided invoice	Not in export set	Negative
TC-28-01	AC-28.1	Primary store encrypted	Docs archived	Inspect stored object	any doc	Encrypted (AES-256)	Positive
TC-28-02	AC-28.2	Backup encrypted	Backup due	Run backup; inspect disk	portable disk	Backup encrypted	Positive
TC-28-03	AC-28.3	Backup unreadable w/o key	Encrypted backup	Mount elsewhere w/o key	foreign machine	Unreadable	Negative
TC-29-01	AC-29.1	Emailed bill captured	Bill saved from inbox	Capture via FR-1	emailed rechnung.pdf	Archived as incoming bill	Positive
TC-22-01	AC-22.1	Verfahrensdok. sections present	Doc delivered	Review vs section list	the document	All required sections present	Positive
TC-22-02	AC-22.2	Advisor sign-off recorded	Doc finalised	Check sign-off	advisor approval	Sign-off present & dated	Positive
TC-22-03	AC-22.3	Adding category updates doc+log	Doc exists	Add category; inspect section+log	new: Permits	Section updated; change logged	Positive

TC-23-01	AC-23.1	Source & capture date shown	Doc archived	Retrieve; read metadata	migrated bill	Source+capture date; addressable	Positive
TC-24-01	AC-24.1	Archived copy byte-identical	PDF captured	Hash source & copy; compare	any PDF	Hashes match	Positive

10. Traceability Matrix

Business Goal	Functional Req	User Story	Test Cases	Status
BG-1	FR-1, FR-6, FR-29	US-1	TC-01-01…10, TC-29-01	✔
BG-1 / BG-2	FR-2, FR-25	US-2	TC-02-01…03, TC-25-01…03	✔
BG-1 / BG-2	FR-3	US-3	TC-03-01…04	✔
BG-2	FR-13, FR-14	US-4	TC-04-01, TC-04-02	✔
BG-1	FR-9…FR-12	US-5	TC-05-01…05	✔
BG-2	FR-15, FR-17, FR-26, FR-28	US-6	TC-06-01…05, TC-26-01…02, TC-28-01…03	✔
BG-2	FR-20, FR-21, FR-22, FR-23, FR-24, FR-27	US-7	TC-07-01…04, TC-22-01…03, TC-23-01, TC-24-01, TC-27-01…04	✔
BG-2	FR-19	US-8	TC-19-01…04	✔

Reverse check: every test traces to an AC; every functional requirement has ≥1 test. No coverage holes. FR-4, FR-5, FR-7, FR-8, FR-18 are supporting/process requirements verified via US-1/US-2 flows and the Verfahrensdokumentation rather than dedicated cases.

11. Open Items & Risks

ID	Item	Owner
RSK-101	€10/mo ceiling vs. GoBD-compliant storage — assumed met by a low-cost cloud DMS TEST; confirm with real pricing.	Owner
RSK-102	Single backup volume is a single point of failure — mitigated by FR-26 restore test; consider a second/offsite copy.	Owner
RSK-103	Manual metadata entry risks inconsistent indexing — mitigated by naming convention in the Verfahrensdokumentation.	Owner
OQ-16	Accounting program is not GoBD-compliant — possible exposure for the accounting records themselves, outside this project.	Tax advisor
OQ-18	GDPR lawful basis & processing record — confirm.	Tax advisor
—	All TEST values (50 MB, AES-256, 15-min session, 5-attempt lockout, cloud DMS choice) require confirmation.	Owner

Produced via the requirements-toolkit workflow: elicitation → drafting → review → test cases, with diagrams. A human owns every final decision; this document is a draft for stakeholder and tax-advisor confirmation.